

Semestertoets 3	Semester Test 3
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Elektrisiteit en Elektronika EBN 122	Electricity and Electronics EBN 122
24 Oktober 2018	24 October 2018

Toetsinligting - Test Information			
Maksimum punte: Maximum marks:	45	Volpunte: Full marks:	40
Duur van vraestel: Duration of paper:	90 min + (10 for OCR)	Oop-/Toeboek: Open/Closed book:	Toe/Closed
Totale aantal bladsye (hierdie blad ingesluit): Total number of pages (this page included)		16	

Important	Belangrik
1. The examination regulations of the University of Pretoria apply. 2. All questions must be answered on the Optical Character Recognition (OCR) form. 3. Read the instructions on the OCR form carefully. 4. Where applicable, answers must include units. 5. Final answers must be written as real numbers and not as fractions. 6. Use 5 significant numbers to write irrational numbers.	1. Die eksamenregulasies van die Universiteit van Pretoria geld. 2. Alle vrae moet op die Optiese Karakter Herkenning (OKH) vorm ingevul word. 3. Neem noukeurig kennis van die instruksies op die OKH vorm. 4. Indien van toepassing, moet antwoorde eenhede insluit. 5. Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie. 6. Gebruik 5 beduidende syfers om irrasionele getalle voor te stel.

Lecturers/Dosente:	D. le Roux, J.P. de Villiers, J.P. Jacobs
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INSTRUCTIONS

Do step 0 in the first 5 minutes of the session. (-5 to 0 min)

Do step 1 in the first 90 minutes of the test. (0-90 min)

Do step 2 within the following 10 minutes. (90-100 min)

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form.

STEP 1 (0 min to 90 min): Do your own calculations on the QUESTION PAPER. The question paper will not be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET will not be taken in at the end of the test.

STEP 2 (90 min to 100 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2018EBN122S3**
- Use a mechanical pencil with 0.5mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

Section 1 / Afdeling 1

Questions 1 to 2 [4]		Vrae 1 tot 2 [4]	
It is known that $v_L(0) = 1$ V and $v_C(0) = 0$ V. Calculate $v_L(t = 1$ s) and $v_C(t = 1$ s).		Dit word gegee dat $v_L(0) = 1$ V en $v_C(0) = 0$ V. Bereken $v_L(t = 1$ s) en $v_C(t = 1$ s).	
1	$v_L(1$ s)	1	$v_L(1$ s)
2	$v_C(1$ s)	2	$v_C(1$ s)

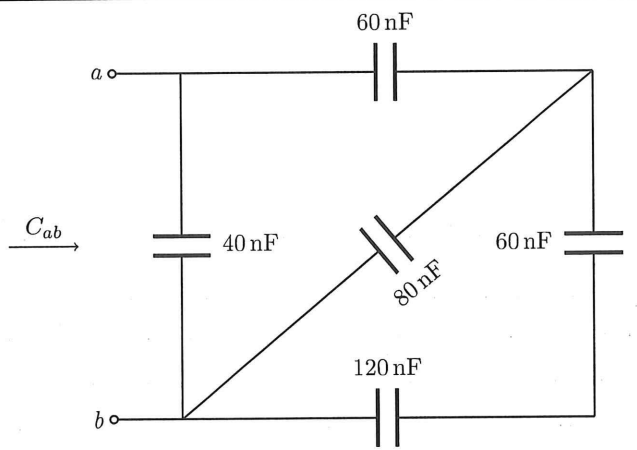
The circuit diagram shows a current source labeled $5 - 5e^{-t}$ A with an upward arrow. This source is in series with a capacitor labeled 5 F and an inductor labeled 0.2 H. The voltage across the capacitor is labeled $v_C(t)$ with the positive terminal at the top. The voltage across the inductor is labeled $v_L(t)$ with the positive terminal at the top.

$$\begin{aligned}
 * \quad v_L &= L \frac{di}{dt} \\
 &= 0.2 \frac{d}{dt}(5 - 5e^{-t}) \\
 &= 0.2(0 + 5e^{-t}) \\
 &= 1e^{-t}
 \end{aligned}$$

$$\begin{aligned}
 \therefore v_L(1) &= e^{-1} \\
 &= \underline{0.3679 \text{ V}} \rightarrow
 \end{aligned}$$

$$\begin{aligned}
 * \quad i_C &= C \frac{dv_C}{dt} \\
 \therefore v_C &= \frac{1}{C} \int_0^t i_C dt + v_C(0) \\
 &= \frac{1}{5} \int_0^t (5 - 5e^{-t}) dt \\
 &= \int_0^t (1 - e^{-t}) dt \\
 &= t + e^{-t} \Big|_0^t \\
 &= t + e^{-t} - 0 - e^{-0} \\
 &= t + e^{-t} - 1.
 \end{aligned}$$

$$\begin{aligned}
 \therefore v_C(1) &= 1 + e^{-1} - 1 \\
 &= \underline{0.3679 \text{ V}} \rightarrow
 \end{aligned}$$

Question 3 [2]		Vraag 3 [2]	
Determine C_{ab} .		Bepaal C_{ab} .	
3	C_{ab}	3	C_{ab}
			

$$C_{ab} = (60 \parallel 120 + 80) \parallel 60 + 40$$

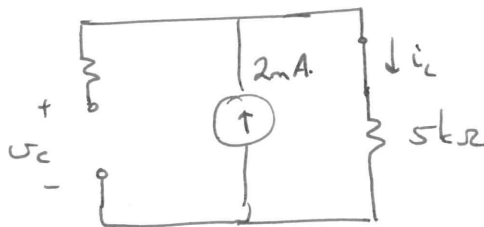
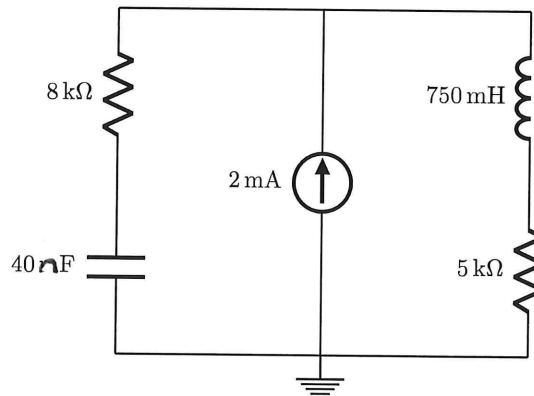
$$= (40 + 80) \parallel 60 + 40$$

$$= 120 \parallel 60 + 40$$

$$= 40 + 40$$

$$= \underline{80 \text{ nF}}$$

Questions 4 to 5 [4]		Vrae 4 tot 5 [4]	
Assume direct-current (steady-state) conditions. Calculate the energy stored in the capacitor, w_C , and inductor, w_L .		Aanvaar gelykstroom (gestadigde) toestand. Bepaal die energie wat in die kapasitor, w_C , en induktor, w_L , gestoor word.	
4	w_L	4	w_L
5	w_C	5	w_C



$$\therefore i_L = 2 \text{ mA} \quad \rightarrow \quad U_C = 2 \text{ mA} \times 5 \text{ k}\Omega = 10 \text{ V}$$

$$w_L = \frac{1}{2} \times 750 \text{ m} \times 2 \text{ m}^2$$

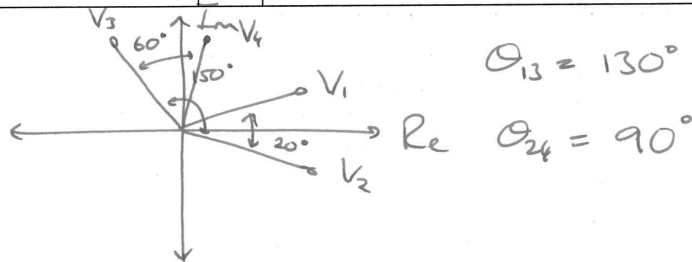
$$= 1.5 \mu\text{J}$$

$$w_C = \frac{1}{2} (40 \text{ n}) 10^2$$

$$= 2 \mu\text{J}$$

Section 2 / Afdeling 2

Questions 6 to 7 [2]		Vrae 6 tot 7 [2]	
$v_1(t)$ leads $v_2(t)$ by 20° . $v_2(t)$ lags $v_3(t)$ by 150° . $v_3(t)$ leads $v_4(t)$ by 60° . Calculate the phase angles: θ_{13} between v_1 and v_3, θ_{24} between v_2 and v_4, such that the angle is between 0° and 180° .		$v_1(t)$ loop voor $v_2(t)$ met 20° . $v_2(t)$ loop na $v_3(t)$ met 150° . $v_3(t)$ loop voor $v_4(t)$ met 60° . Bepaal die fasehoeke: θ_{13} tussen v_1 en v_3, θ_{24} tussen v_2 en v_4, sodat die fasehoek tussen 0° en 180° is.	
6	θ_{13}	6	θ_{13}
7	θ_{24}	7	θ_{24}



Question 8 [2]		Vraag 8 [2]	
Solve for x and write the answer in either polar or rectangular form. $\frac{x + 6\angle 0^\circ}{12 + j8} + \frac{x}{j4} = 1\angle 90^\circ$		Los op vir x en skryf die antwoord in of polêre of reghoekige formaat. $\frac{x + 6\angle 0^\circ}{12 + j8} + \frac{x}{j4} = 1\angle 90^\circ$	
8	x	8	x

$$\frac{x+6}{12+j8} + \frac{x}{j4} = j$$

$$(x+6)j4 + (12+j8)x = j(j4)(12+j8)$$

$$j4x + j24 + 12x + j8x = -48 - j32$$

$$x(12+j12) = -48 - j32 - j24$$

$$x = \frac{-48 - j56}{12+j12}$$

$$= -4.333 - j0.333$$

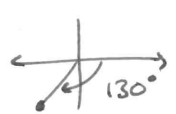
$$= 4.346 \angle -175.6013^\circ$$

Question 9 [2]	Vraag 9 [2]
The following sinusoids are given: $i_1(t) = -4 \cos(\omega t - 130^\circ) \text{ A}$ $i_2(t) = -3 \sin(\omega t + 140^\circ) \text{ A}$ Determine the phasor value of $\bar{I}_T = \bar{I}_1 + \bar{I}_2$. Give the answer in polar form.	Die volgende sinusiede word gegee: $i_1(t) = -4 \cos(\omega t - 130^\circ) \text{ A}$ $i_2(t) = -3 \sin(\omega t + 140^\circ) \text{ A}$ Bepaal die fasor waarde van $\bar{I}_T = \bar{I}_1 + \bar{I}_2$. Gee die antwoord in polêre vorm.
9 $\bar{I}_T$	9 $\bar{I}_T$



$$i_1 = -4 \cos(\omega t - 130^\circ)$$

$$= 4 \cos(\omega t + 50^\circ) \Rightarrow I_1 = 4 \angle 50^\circ \text{ A}$$



$$i_2 = -3 \sin(\omega t + 140^\circ)$$

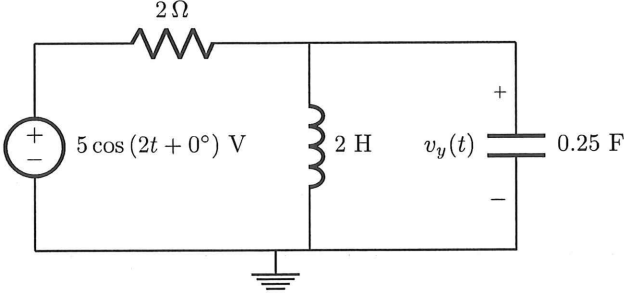
$$= 3 \cos(\omega t - 130^\circ) \Rightarrow I_2 = 3 \angle -130^\circ \text{ A}$$

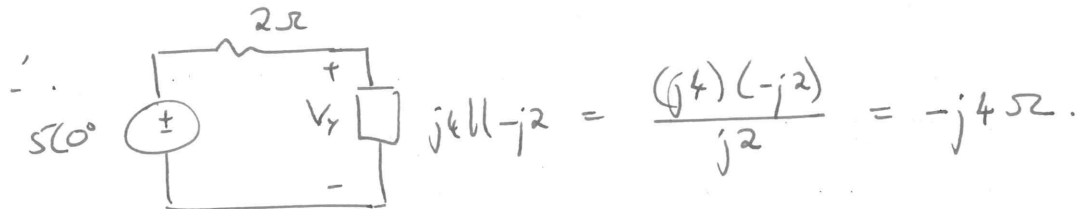
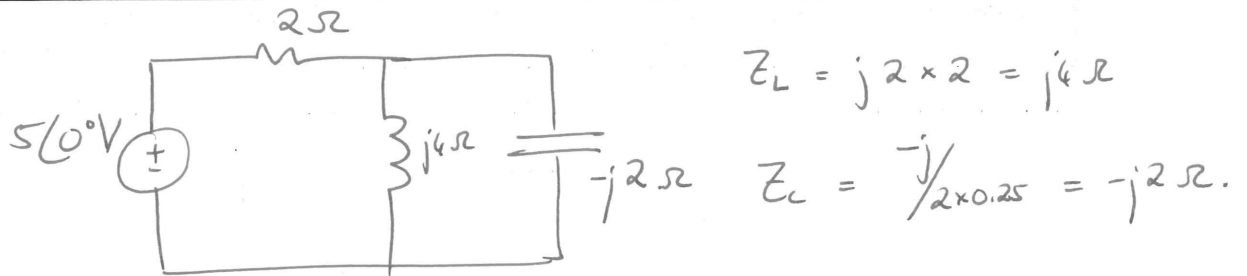
$$I_T = 1 \angle 50^\circ \text{ A}$$

Question 10 [2]	Vraag 10 [2]
A parallel circuit has a $0.2 \cos(5 \times 10^3 t)$ A current source, a 40Ω resistor, and a 24 mH inductor. Calculate the voltage $\bar{V}_T$ across the current source. The voltage is measured as positive where current exits the source and negative where the current enters the source. Give the answer in rectangular form.	'n Parallel-kringbaan bestaan uit 'n $0.2 \cos(5 \times 10^3 t)$ A stroombron, 'n 40Ω resistor, en 'n 24 mH induktor. Bepaal die spanning $\bar{V}_T$ oor die stroombron. Die spanning word gemeet as positief waar stroom uit die bron vloeit, en negatief waar die stroom in die bron vloeit. Gee die antwoord in reghoekige vorm.
10 $\bar{V}_T$	10 $\bar{V}_T$



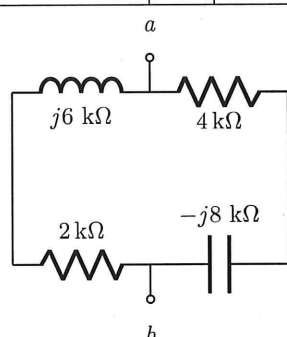
$$\begin{aligned}
 V_T &= 0.2 \times (40 \parallel j120) \\
 &= \frac{0.2 \times 40 \times (j120)}{40 + j120} \\
 &= \underline{7.2 + j2.4 \text{ V}}
 \end{aligned}$$

Question 11 [2]		Vraag 11 [2]	
Calculate the phasor voltage $\bar{V}_Y$. Give the answer in polar form .		Bereken die fasorspanning $\bar{V}_Y$. Gee die antwoord in polêre vorm .	
11	$\bar{V}_Y$	11	$\bar{V}_Y$
			

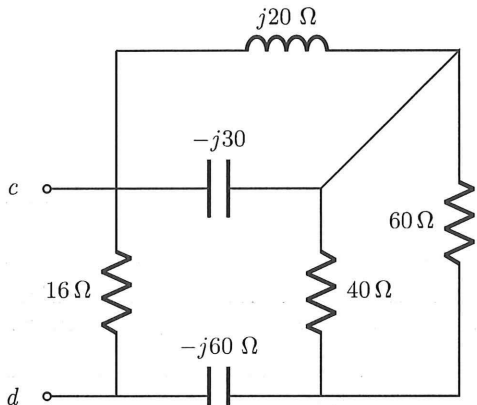


$$V_Y = \frac{5 \times (-j4)}{2 - j4} = \frac{4 - j2}{2 - j4} \text{ V}$$

$$= 4.47 \angle -26.57^\circ \text{ V}$$

Question 12 [2]		Vraag 12 [2]	
Determine Z_{ab} . Give the answer in rectangular form .		Bepaal Z_{ab} . Gee die antwoord in reghoekige vorm .	
12	Z_{ab}	12	Z_{ab}
			

$$\begin{aligned}
 Z_{ab} &= (2 + j6) \parallel (4 - j8) \times 10^3 \\
 &= \frac{(2 + j6)(4 - j8) \times 10^3}{(6 - j2)} \\
 &= \underline{8 + j4 \text{ k}\Omega}
 \end{aligned}$$

Question 13 [2]		Vraag 13 [2]	
Determine Z_{cd} . Give the answer in rectangular form .		Bepaal Z_{cd} . Gee die antwoord in reghoekige vorm .	
13	Z_{cd}	13	Z_{cd}
			

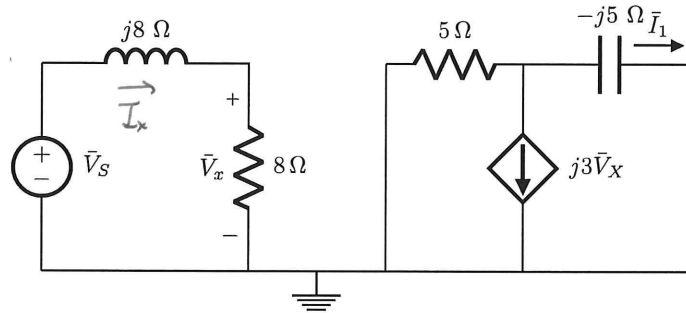
$$Z_{cd} = 16 \parallel (-j60 + 40 \parallel 60 + j20 \parallel (-j30))$$

$$= 16 \parallel (-j60 + 24 + j60)$$

$$= 16 \parallel 24$$

$$= \underline{9.6 \Omega}$$

Questions 14 to 15 [4]		Vrae 14 tot 15 [4]	
Given: $\bar{V}_x = -j2V$. Calculate the source voltage $\bar{V}_S$ and the current $\bar{I}_1$. Give the answers in rectangular form.		Gegee: $\bar{V}_x = -j2V$. Bepaal die bronspanning $\bar{V}_S$ en die stroom $\bar{I}_1$. Gee die antwoorde in reghoekige vorm.	
14	$\bar{V}_S$	14	$\bar{V}_S$
15	$\bar{I}_1$	15	$\bar{I}_1$



KVL: $-V_S + j8I_x + 8I_x = 0$

$$\begin{aligned}
 V_S &= (8 + j8)I_x = (8 + j8) \frac{V_x}{8} \\
 &= (1 + j)V_x = (1 + j)(-j2) \\
 &= 2 - j2 \text{ V}
 \end{aligned}$$

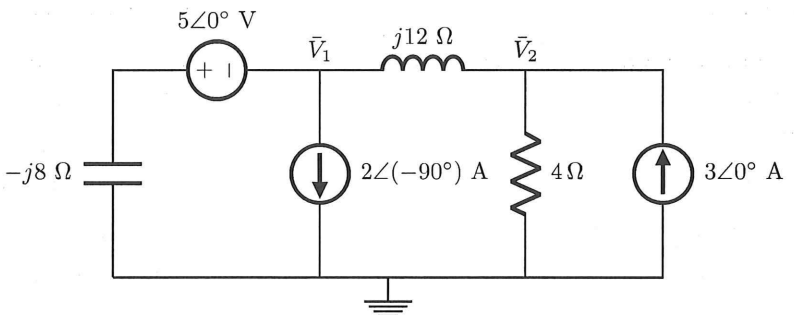
Current division:

$$\begin{aligned}
 I_1 &= \frac{-(j3V_x)(5)}{5 - j5} \\
 &= \frac{(-j3)(-j2)}{1 - j} \\
 &= -3 - j3 \text{ A}
 \end{aligned}$$

* Voltage division: $V_x = \frac{V_S \times 8}{8 + j8} = V_S / (1 + j)$

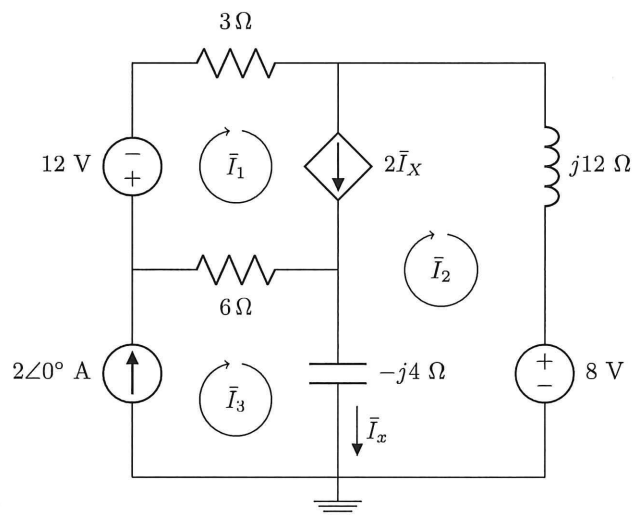
$\therefore V_S = V_x (1 + j) = 2 - j2 \text{ V.}$

Section 3 / Afdeling 3

Questions 16 to 19 [4]				Vrae 16 tot 19 [4]			
Use nodal analysis to set up the following two equations of the form: $a\bar{V}_1 + b\bar{V}_2 = 33$ $c\bar{V}_1 + d\bar{V}_2 = j36$ Calculate the unknown coefficients and write in rectangular format.				Gebruik nodale analise om die volgende twee vergelykings op te stel met die vorm: $a\bar{V}_1 + b\bar{V}_2 = 33$ $c\bar{V}_1 + d\bar{V}_2 = j36$ Bepaal die onbekende koëffisiënte en skryf in reghoekige formaat.			
16	a [1]	17	b [1]	18	c [1]	19	d [1]
							

$$\begin{aligned}
 * \quad -j2 + \frac{V_1 + 5}{-j8} + \frac{V_1 - V_2}{j12} &= 0 \\
 (x j24) \quad (-j2)(j24) - 3V_1 - 15 + 2V_1 - 2V_2 &= 0. \\
 -V_1 - 2V_2 &= 15 + j^2 48 \\
 -V_1 - 2V_2 &= -33 \\
 V_1 + 2V_2 &= 33 \quad \underline{\hspace{1cm}} \quad 2 \\
 * \quad 3 &= \frac{V_2}{4} + \frac{V_2 - V_1}{j12} \\
 (x j12) \quad j36 &= j3V_2 + V_2 - V_1 \\
 -V_1 + (1 + j3)V_2 &= j36 \quad \underline{\hspace{1cm}} \quad 2
 \end{aligned}$$

Questions 20 to 23 [4]				Vrae 20 tot 23 [4]			
Use mesh analysis to set up the following two equations of the form:				Gebruik lusanalise om die volgende twee vergelykings op te stel met die vorm:			
$e\bar{I}_1 + f\bar{I}_2 = -8 - j8$ $g\bar{I}_1 + h\bar{I}_2 = 4$				$e\bar{I}_1 + f\bar{I}_2 = -8 - j8$ $g\bar{I}_1 + h\bar{I}_2 = 4$			
Calculate the unknown coefficients and write in rectangular format.				Bepaal die onbekende koëffisiënte en skryf in reghoekige formaat.			
20	e [1]	21	f [1]	22	g [1]	23	h [1]



Supermesh *

$$+12 + 3I_1 + j12I_2 + 8 - j4I_2 + j4I_3 + 6I_1 - 6I_3 = 0$$

$$20 + 9I_1 + j8I_2 - 6I_3 + j4I_3 = 0$$

$$9I_1 + j8I_2 = -20 + 6(2) - j4(2)$$

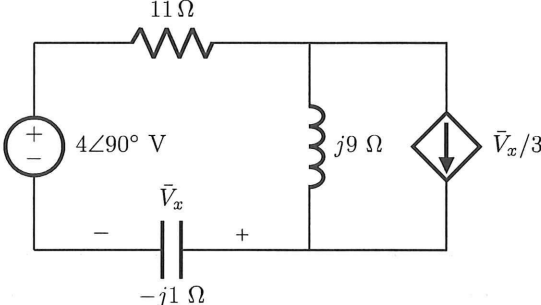
$$9I_1 + j8I_2 = -8 - j8$$

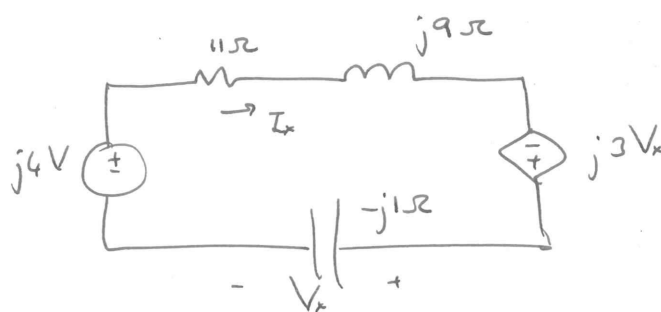
* $I_x = I_3 - I_2 = 2 - I_2$ (Bottom node)

* $2I_x = I_1 - I_2$ (CCCS)

$$2(2 - I_2) = I_1 - I_2$$

$$4 = I_1 + I_2$$

Question 24 [2]	Vraag 24 [2]
Use source transformation and determine $\bar{V}_x$. Write the answer in polar form .	Gebruik brontransformasie en bepaal $\bar{V}_x$. Skryf die antwoord in polêre vorm .
24 $\bar{V}_x$	24 $\bar{V}_x$
	



$$\rightarrow -j4 + 11I_x + j9I_x - j3V_x - jI_x = 0.$$

$$(11 + j8)I_x - j3V_x = j4$$

$$\text{but } I_x = \frac{V_x}{-j}$$

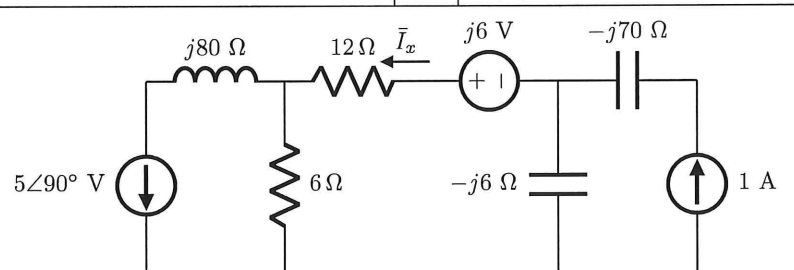
$$\therefore \frac{(11 + j8)V_x}{-j} - j3V_x = j4$$

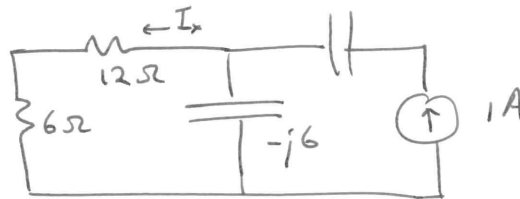
$$(j11 - 8 - j3)V_x = j4$$

$$V_x = \frac{j4}{-8 + j8}$$

$$= 0.25 - j0.25 \text{ V}$$

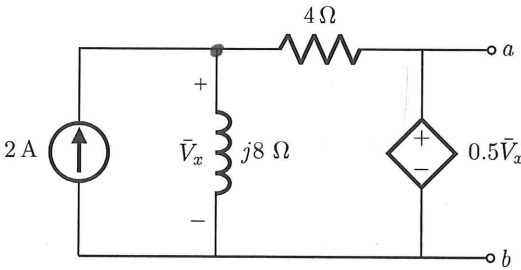
$$\boxed{V_X = 0.354 \angle -45^\circ \text{ V}}$$

Question 25 [2]		Vraag 25 [2]	
Calculate the exclusive contribution of the 1 A current source to $\bar{I}_x$. Write the answer in polar form .		Bepaal die bydra van alleenlik die 1 A stroombron tot $\bar{I}_x$. Skryf die antwoord in polêre vorm .	
25	$\bar{I}_{x(1A)}$	25	$\bar{I}_{x(1A)}$
			



$$\begin{aligned}
 \bar{I}_x &= \frac{1 \times (-j6)}{18 - j6} \\
 &= \underline{0.1 - j0.3 \text{ A}}
 \end{aligned}$$

$$\bar{I}_X = 0.316 \angle -71.57^\circ \text{ A}$$

Question 26 [2]	Vraag 26 [2]
Determine the Thevenin equivalent voltage for the circuit below with respect to terminals a - b . Give the answers in rectangular form .	Bepaal die Thevenin ekwivalente spanning vir onderstaande kringbaan met verwysing tot terminale a - b . Gee die antwoorde in reghoekige vorm .
26 $\bar{V}_{TH}$	26 $\bar{V}_{TH}$
	

$$\bar{V}_{TH}$$

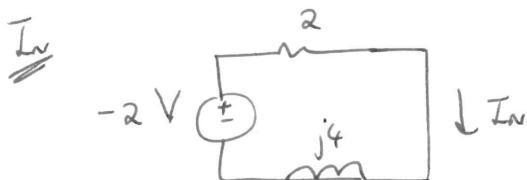
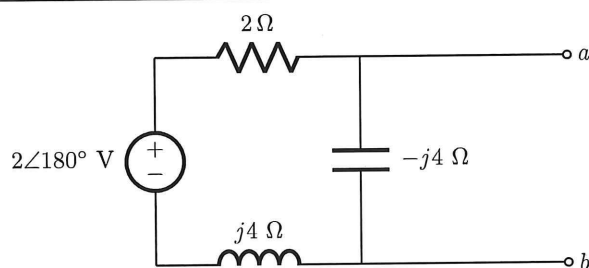
$$2 = \frac{V_x}{j8} + \frac{V_x - 0.5V_x}{4}$$

$$(xj8) \quad j16 = V_x + 0.5V_x(j2)$$

$$V_x = \frac{j16}{1+j} = 8 + j8 \text{ V}$$

$$\therefore \bar{V}_{TH} = 0.5(8 + j8) \\ = \underline{4 + j4 \text{ V}}$$

Questions 27 to 28 [4]		Vrae 27 tot 28 [4]	
Determine the Norton equivalent current and impedance for the circuit below with respect to terminals a - b . Give the answers in rectangular form .		Bepaal die Norton ekwivalente stroom en impedansie vir onderstaande kringbaan met verwysing tot terminale a - b . Gee die antwoorde in reghoekige vorm .	
27	$\bar{I}_N$	27	$\bar{I}_N$
28	Z_N	28	Z_N



$$\bar{I}_N = \frac{-2}{2 + j4}$$

$$= -0.2 + j0.4 \text{ A}$$



$$Z_N = (2 + j4) \parallel (-j4)$$

$$= \frac{(2 + j4)(-j4)}{2}$$

$$= -j4 + 8$$

$$= 8 - j4 \text{ } \Omega$$

Total Section 3 / Totaal Afdeling 3: [18]

Grand Total / Groot totaal : [46]